

Digital Clock in Python (Tkinter) - Complete Explanation

This document explains the Digital Clock project created using Python and Tkinter. The project displays the current time and date in a graphical window and updates automatically every second.

Corrected Code

```
import tkinter as tk
import time
root = tk.Tk()
root.title("Digital Clock")
def update_time():
    string = time.strftime('%H:%M:%S %p \n %D')
    label.config(text=string)
    label.after(1000, update_time)
label = tk.Label(root, font=('calibri', 40, 'bold'), background='purple')
label.pack(anchor='center')
update_time()
root.mainloop()
```

1. import tkinter as tk

Tkinter is Python's built-in GUI library. It is used to create windows, labels, buttons, and other graphical components.

2. import time

The time module provides functions related to time. Here it is used to get the current system time using `strftime()`.

3. root = tk.Tk()

Creates the main application window.

4. root.title('Digital Clock')

Sets the title displayed on the window's title bar.

5. def update_time():

Defines a function that updates the displayed time every second.

6. time.strftime()

Formats the current time. %H = hour, %M = minute, %S = second, %p = AM/PM, %D = date.

7. label.config(text=string)

Updates the label text with the latest time and date.

8. label.after(1000, update_time)

Runs `update_time()` again after 1000 milliseconds (1 second), creating a live clock.

9. tk.Label()

Creates a label widget used to display the time on the window.

10. label.pack(anchor='center')

Places the label in the application window.

11. update_time()

Starts the clock updating process.

12. root.mainloop()

Starts Tkinter's event loop and keeps the window running.

Common Mistake in Your Screenshot

The line `'import strftime'` causes a warning because `strftime` is not a separate Python module. It is a function inside the `time` module. Use `time.strftime()` instead.